Detailed derivation of free–free absorption by electron–atom collisions at terahertz frequencies

Cooper K. Watson

October 20, 2026

These notes derive, with every intermediate result written out, the rate at which an electromagnetic wave of terahertz frequency is absorbed by free electrons that collide with neutral atoms. They ask whether the classical absorption kernel, the Lorentz-gas conductivity built on the momentum-transfer cross section, is accurate at the terahertz frequencies proposed for reheating the electrons of an electron-beam-pumped KrF laser gas. The electron is described by exact scattering states of a local potential that represents the atom, the field is treated to first order, and the rate follows from the golden rule. Every boxed result is checked numerically, as Appendix C describes.

For notation, Gaussian units are used. The electron has charge $-e$ and mass m, and the atom is infinitely heavy and sits at the origin. A state of momentum $\hbar\mathbf{k}$ has energy $E = \hbar^2k^2/2m$ and speed $v = \hbar k/m$. The wave is uniform and linearly polarized along the unit vector $\hat{\mathbf{e}}$, with amplitude E_0 and angular frequency ω, and $f = \omega/2\pi$ is its frequency. The absorption of one photon takes the electron from $\hbar\mathbf{k}$ to $\hbar\mathbf{k}'$ with $k'^2 = k^2 + 2m\omega/\hbar$. The temperature of a Maxwell distribution is T and $u = \hbar\omega/k_BT$. The momentum-transfer cross section is $\sigma_{\mathrm{m}}(E)$, and a cross section with a subscript $+$ or $-$ is the free–free cross section for absorption or stimulated emission per unit photon flux, defined in Sec. 3. The numbers are computed in atomic units, $\hbar = m = e = 1$, with the bohr a_0 and the hartree $E_h = 27.211$ eV. A boxed equation is a result that is checked.

Contents

1 Why the atoms matter **2**

2 The Hamiltonian **3**
2.1 One electron and one atom . . . 3
2.2 The wave . . . 4
2.3 The response of the atom to the wave . . . 4

3 The golden-rule cross section between exact scattering states **4**
3.1 The scattering states . . . 4
3.2 The matrix element . . . 5
3.3 The rate and its average over directions . . . 5
3.4 The cross section per photon flux . . . 6
3.5 Length, velocity and acceleration forms . . . 6

4 Emission, detailed balance and the absorption coefficient **6**
4.1 Stimulated emission and detailed balance . . . 6
4.2 The absorption coefficient . . . 7
4.3 The kinetic equation and its Fokker–Planck limit . . . 7

5 The soft-photon limit **8**
5.1 The force identity . . . 8
5.2 The soft-photon cross section . . . 9
5.3 The classical limit is the Lorentz gas . . . 9
5.4 The free–free Gaunt factor and the error of the classical kernel . . . 10

6 Exact limits at finite photon energy **10**
6.1 The zero-range potential . . . 10
6.2 The Born approximation . . . 11
6.3 The first order in the photon energy . . . 11

7 Strong fields and the weak-field criterion **11**

8 Realistic atoms **12**
8.1 Three families of model potentials . . . 12
8.2 Numerical method . . . 14
8.3 The free–free Gaunt factor . . . 14
8.4 Polarization bremsstrahlung . . . 15

9 Thermal absorption and the error of the classical kernel **15**
9.1 The pure gases . . . 15
9.2 The two fills . . . 15
9.3 The answer for the laser gas . . . 16

10 Electron–ion or electron–atom **16**

11 Summary **17**

A Partial waves **19**

B The integrals of the zero-range limit **19**

C The numerical checks **20**

1 Why the atoms matter

(1) Inverse bremsstrahlung, the absorption of a wave by electrons that scatter on ions, heats laser plasmas [1]. In the gas of an electron-beam-pumped KrF laser the electron density is 10^{19} to 2.7×10^{21} m^{-3} against an atom density of 2.4×10^{25} to 4.9×10^{25} m^{-3}, so that there are 9×10^{3} to 5×10^{6} atoms for every ion. The Coulomb cross section is larger by orders of magnitude, however, and Sec. 10 finds the regime in which each kind of collision dominates the absorption. Away from the coldest and densest afterglow the atoms win.
(2) Free–free absorption on neutral atoms has been computed from phase shifts for the negative hydrogen ion [11] and for other atoms [12], and with the multichannel R-matrix method for argon

[13]. Its inverse, the neutral bremsstrahlung, has been measured [14, 15] and explains part of the electroluminescence of two-phase argon detectors [16].
(3) Proposals to reheat the electrons of the laser gas with a terahertz wave compute its absorption length with the classical kernel of the Lorentz gas. That kernel is the conductivity of electrons that scatter off heavy atoms with the energy dependent collision frequency $\nu(v) = N\sigma_{\rm m}(v)v$, derived in Sec. 5.3. It treats the electron as a classical particle that absorbs energy continuously. Two quantum effects are missing from it. The first is the ratio $u = \hbar\omega/k_BT$, which is 0.16 at 1 THz and 4.5 at the 28.3 THz of a CO_2 laser for electrons at the gas temperature of 300 K. The second is the structure of $\sigma_{\rm m}(E)$. Argon and krypton have a Ramsauer–Townsend window in which $\sigma_{\rm m}$ falls from its value at 1 meV by a factor of 82 in argon and 330 in krypton, to minima at 0.25 and 0.51 eV in the LXCat sets, and a photon of 28.3 THz carries 0.117 eV. Table 1 lists the photon energy and u at the frequencies used below.

Table 1: Photon energy $\hbar\omega$, the ratio $u = \hbar\omega/k_BT$ and the stimulated-emission factor $(1 - e^{-u})/u$ at 300 K, and u for electrons at 0.63 eV.

f (THz)	$\hbar\omega$ (meV)	u, 300 K	$(1 - e^{-u})/u$, 300 K	u, 0.63 eV
0.3	1.24	0.048	0.976	0.00197
1	4.14	0.16	0.924	0.00656
3	12.4	0.48	0.794	0.0197
10	41.4	1.6	0.499	0.0656
28.3	117	4.53	0.218	0.186

(4) The question is the error of the classical kernel at 3 to 28 THz in two fills of the laser gas, $Ne/Kr/F_2 = 0.948/0.05/0.002$ at 2 atm and $Ar/Kr/F_2 = 0.597/0.40/0.003$ at 1 atm. The kernel is good enough if the error is below 5 % in the band of 0.3 to 4 THz proposed for reheating. Sections 2 to 4 derive the quantum free–free cross section of one electron and one atom exactly within a one-body model. Section 5 finds its soft-photon and classical limits, and Sec. 6 two limits in which it is known in closed form at any photon energy. Sections 8 and 9 evaluate it for model potentials fitted to the measured cross sections of neon, argon and krypton. The answer is in Sec. 9 and in the Summary.

2 The Hamiltonian

2.1 One electron and one atom

(1) The atom is represented by a local central potential $V(r)$ that acts on the scattering electron,

$$H_0 = \frac{p^2}{2m} + V(r), \qquad V(r) \to -\frac{\alpha_d e^2}{2r^4} \quad (r \to \infty), \tag{2.1}$$

where α_d is the static dipole polarizability of the atom. At short range V contains the screened attraction of the nucleus and the repulsion that the Pauli principle produces. In the one-body model the latter appears because the occupied orbitals of the atom are bound states of the same potential, so that the scattering states are orthogonal to them. The models of Sec. 8 are of this kind.

(2) The atom is infinitely heavy. Its recoil changes the electron energy by a fraction of order m/M, which is 1.4×10^{-5} for argon, and is neglected. Exchange between the scattering electron and the atomic electrons, and the excitation of the atom, are outside the model. The second needs at least 11.5 eV in argon and 9.9 eV in krypton and is closed for the electrons that absorb terahertz photons.

2.2 The wave

(1) The wave has the field $\mathbf{E}(t) = E_0 \hat{\mathbf{e}} \cos \omega t$. At 1 THz its wavelength is 0.3 mm and at 28.3 THz it is 10.6 μm. A collision takes place within a few hundred a_0 of the atom, so the field is uniform over it and the dipole approximation holds. In the length gauge the potential energy of the charge $-e$ in the field is $e\mathbf{E}(t) \cdot \mathbf{r}$, and

$$H(t) = H_0 + eE_0 \cos(\omega t)\, \hat{\mathbf{e}} \cdot \mathbf{r} = H_0 + \frac{eE_0}{2}\, \hat{\mathbf{e}} \cdot \mathbf{r} \left(e^{-i\omega t} + e^{i\omega t}\right). \tag{2.2}$$

The term with $e^{-i\omega t}$ absorbs a photon and the term with $e^{i\omega t}$ emits one.
(2) The field is weak when the quiver amplitude $a = eE_0/m\omega^2$ is small against the wavelength of the electron. The criterion and its numbers are in Sec. 7. In the weak field the golden rule to first order in E_0 describes the absorption of one photon.

2.3 The response of the atom to the wave

(1) The wave also polarizes the atom. Its induced dipole is $\alpha_d \mathbf{E}(t)$, since $\hbar\omega$ is far below the excitation energies of the atom. The electrostatic potential of a dipole $\mathbf{d}$ at the origin is $\mathbf{d} \cdot \mathbf{r}/r^3$, so the electron acquires the potential energy $-e\alpha_d \mathbf{E}(t) \cdot \mathbf{r}/r^3$. Together with Eq. (2.2) the interaction with the wave becomes

$$e\mathbf{E}(t) \cdot \mathbf{r} \Big(1 - \frac{\alpha_d}{r^3}\Big). \tag{2.3}$$

The second term is the polarization bremsstrahlung of the atom, the photon absorbed through the dipole of the atom rather than through the electron. Inside the atom the field of the dipole is screened, so the term is cut off there. Section 8.4 computes its effect with the same cut-off as the polarization potential, and finds it below 0.7 % up to 3 THz. Until then it is left out.

3 The golden-rule cross section between exact scattering states

3.1 The scattering states

(1) The scattering states of H_0 with energy $E = \hbar^2 k^2/2m$ are normalized to $\langle \psi_{\mathbf{k}} | \psi_{\mathbf{k}'} \rangle = \delta^3(\mathbf{k} - \mathbf{k}')$. In partial waves,

$$\psi_{\mathbf{k}}^{(\pm)}(\mathbf{r}) = (2\pi)^{-3/2} \sum_{l=0}^{\infty} (2l+1)\, i^l e^{\pm i\delta_l}\, \frac{u_{kl}(r)}{kr}\, P_l(\hat{\mathbf{k}} \cdot \hat{\mathbf{r}}), \tag{3.1}$$

where u_{kl} is the real regular solution of the radial equation

$$u_{kl}'' + \Big[k^2 - U(r) - \frac{l(l+1)}{r^2}\Big] u_{kl} = 0, \qquad U(r) = \frac{2mV(r)}{\hbar^2}, \tag{3.2}$$

normalized to unit amplitude, $u_{kl} \to \sin(kr - l\pi/2 + \delta_l)$ as $r \to \infty$, and $\delta_l(k)$ is the phase shift. For $V = 0$ the radial function is the Riccati–Bessel function $\hat{\jmath}_l(kr) = kr\, j_l(kr)$, and Eq. (3.1) is the expansion of the plane wave $(2\pi)^{-3/2} e^{i\mathbf{k}\cdot\mathbf{r}}$. The state $\psi^{(+)}$ has an outgoing scattered wave and is

the initial state. The state $\psi^{(-)}$, with an incoming scattered wave, is the final state of an electron detected in the direction $\hat{\mathbf{k}}'$.
(2) With $P_l(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}) = \frac{4\pi}{2l+1}\sum_m Y^*_{lm}(\hat{\mathbf{k}})Y_{lm}(\hat{\mathbf{r}})$,

$$\psi^{(\pm)}_{\mathbf{k}}(\mathbf{r}) = (2\pi)^{-3/2}\,\frac{4\pi}{kr}\sum_{lm} i^l e^{\pm i\delta_l} u_{kl}(r)\,Y^*_{lm}(\hat{\mathbf{k}})\,Y_{lm}(\hat{\mathbf{r}}). \tag{3.3}$$

3.2 The matrix element

(1) The absorption term of Eq. (2.2) has the matrix element $(eE_0/2)M$ with $M = \langle\psi^{(-)}_{\mathbf{k}'}|\hat{\mathbf{e}}\cdot\mathbf{r}|\psi^{(+)}_{\mathbf{k}}\rangle$. The complex conjugate of Eq. (3.3) for the final state has the factors $(-i)^{l'}e^{+i\delta'_{l'}}Y_{l'm'}(\hat{\mathbf{k}}')Y^*_{l'm'}(\hat{\mathbf{r}})$, with $\delta'_{l'} = \delta_{l'}(k')$. The factor r^2 of d^3r cancels the two factors $1/r$, and

$$M = \frac{(4\pi)^2}{(2\pi)^3 kk'}\sum_{lm,l'm'}(-i)^{l'}i^l e^{i(\delta_l+\delta'_{l'})}\,Y_{l'm'}(\hat{\mathbf{k}}')Y^*_{lm}(\hat{\mathbf{k}})\,\langle l'm'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle\,R_{l\to l'}, \tag{3.4}$$

with the angular matrix element $\langle l'm'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle = \int Y^*_{l'm'}\,\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}\,Y_{lm}\,\mathrm{d}\Omega$ and the radial integral

$$R_{l\to l'} = \int_0^\infty u_{k'l'}(r)\,r\,u_{kl}(r)\,\mathrm{d}r. \tag{3.5}$$

The integrand grows like r at large r, and $R_{l\to l'}$ is defined with the convergence factor $e^{-\eta r}$ and $\eta\to 0^+$. Section 3.5 replaces it by an absolutely convergent integral.
(2) The angular matrix element vanishes unless $l' = l\pm 1$, the dipole selection rule. Appendix A sums its square over m and m',

$$\sum_{mm'}|\langle l+1,m'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle|^2 = \frac{l+1}{3}, \qquad \sum_{mm'}|\langle l-1,m'|\hat{\mathbf{e}}\cdot\hat{\mathbf{r}}|lm\rangle|^2 = \frac{l}{3}. \tag{3.6}$$

3.3 The rate and its average over directions

(1) The golden rule gives the rate of absorption from the state $\psi^{(+)}_{\mathbf{k}}$ into all final directions,

$$W_+ = \frac{2\pi}{\hbar}\Big(\frac{eE_0}{2}\Big)^2\int \mathrm{d}^3k'\,|M|^2\,\delta\Big(\frac{\hbar^2k'^2}{2m} - \frac{\hbar^2k^2}{2m} - \hbar\omega\Big). \tag{3.7}$$

(2) The electrons of a gas move in all directions, so W_+ is averaged over $\hat{\mathbf{k}}$. The integral over $\hat{\mathbf{k}}'$ and the average over $\hat{\mathbf{k}}$ use the orthonormality of the Y_{lm}, $\int Y_{l'm'}(\hat{\mathbf{k}}')Y^*_{l''m''}(\hat{\mathbf{k}}')\mathrm{d}\Omega' = \delta_{l'l''}\delta_{m'm''}$ and $\frac{1}{4\pi}\int Y^*_{lm}(\hat{\mathbf{k}})Y_{\tilde{l}\tilde{m}}(\hat{\mathbf{k}})\mathrm{d}\Omega = \delta_{l\tilde{l}}\delta_{m\tilde{m}}/4\pi$. The cross terms between different pairs (l,l') vanish, the phases drop out, and with Eq. (3.6)

$$\frac{1}{4\pi}\int\mathrm{d}\Omega\int\mathrm{d}\Omega'\,|M|^2 = \frac{(4\pi)^4}{(2\pi)^6k^2k'^2}\,\frac{1}{4\pi}\,\frac{S}{3} = \frac{S}{3\pi^3k^2k'^2}, \tag{3.8}$$

$$S = \sum_{l=0}^{\infty}\Big[(l+1)\,R^2_{l\to l+1} + l\,R^2_{l\to l-1}\Big]. \tag{3.9}$$

(3) The energy delta function contributes $\int k'^2\mathrm{d}k'\,\delta(\hbar^2k'^2/2m - E - \hbar\omega) = mk'/\hbar^2$. The averaged rate is

$$\langle W_+\rangle = \frac{2\pi}{\hbar}\,\frac{e^2E_0^2}{4}\,\frac{mk'}{\hbar^2}\,\frac{S}{3\pi^3k^2k'^2} = \frac{e^2E_0^2\,m\,S}{6\pi^2\hbar^3k^2k'}. \tag{3.10}$$

3.4 The cross section per photon flux

(1) The state $\psi^{(+)}_{\mathbf{k}}$ carries the incident flux $(2\pi)^{-3}\hbar k/m$ of electrons, and the wave carries the flux $\Phi = I/\hbar\omega$ of photons with $I = cE_0^2/8\pi$. The free–free cross section for absorption is the rate divided by both fluxes, $\sigma_+ = \langle W_+\rangle/[(2\pi)^{-3}(\hbar k/m)\Phi]$. With $(2\pi)^3 \cdot 8\pi/6\pi^2 = 32\pi^2/3$,

$$\boxed{\sigma_+(k \to k') = \frac{32\pi^2}{3}\,\frac{e^2m^2\omega}{\hbar^3 c}\,\frac{S}{k^3k'}, \qquad \hbar k' = (\hbar^2k^2 + 2m\hbar\omega)^{1/2}.} \tag{3.11}$$

It has the dimension length$^4\times$time. An electron of speed v among N atoms per unit volume, in the photon flux Φ, absorbs at the rate $N\Phi v\sigma_+$.
(2) In the first Born approximation the radial functions are $\hat{\jmath}_l$ and Eq. (3.11) must equal the one-photon limit of the cross section of Bunkin and Fedorov [4], with $J_1^2(x) \simeq x^2/4$, divided by Φ. The check script compares the two for a Gaussian potential and finds agreement to 10^{-14}, which tests the factor $32\pi^2/3$ and the angular sums together.

3.5 Length, velocity and acceleration forms

(1) For eigenstates of H_0 the commutators $[H_0, \mathbf{r}] = -i\hbar\mathbf{p}/m$ and $[H_0, \mathbf{p}] = i\hbar\nabla V$ give $[H_0, [H_0, \mathbf{r}]] = (\hbar^2/m)\nabla V$. Between states that differ in energy by $\hbar\omega$ the double commutator is $(\hbar\omega)^2$ times the matrix element of $\mathbf{r}$, so

$$\langle f|\mathbf{r}|i\rangle = \frac{-i}{m\omega}\langle f|\mathbf{p}|i\rangle = \frac{1}{m\omega^2}\langle f|\nabla V|i\rangle. \tag{3.12}$$

For scattering states the boundary terms at infinity oscillate and vanish with the convergence factor $e^{-\eta r}$, so the identities hold for the regularized integrals of Eq. (3.5).
(2) Since $\nabla V = V'(r)\hat{\mathbf{r}}$ has the same angular structure as $\mathbf{r} = r\hat{\mathbf{r}}$, the identity holds partial wave by partial wave,

$$\boxed{R_{l\to l'} = \frac{1}{m\omega^2}\int_0^\infty u_{k'l'}(r)\,V'(r)\,u_{kl}(r)\,\mathrm{d}r.} \tag{3.13}$$

The acceleration form converges absolutely when V' falls faster than $1/r$, as the r^{-5} of the polarization tail does, and the numerics use it. The form makes plain that the photon is absorbed where the electron is accelerated, near the atom. The check script verifies Eq. (3.13) against the closed form of the length integral in the zero-range limit of Sec. 6.1.

4 Emission, detailed balance and the absorption coefficient

4.1 Stimulated emission and detailed balance

(1) The term with $e^{i\omega t}$ in Eq. (2.2) takes an electron from $\hbar\mathbf{k}'$ to $\hbar\mathbf{k}$ and adds a photon to the wave. The same steps give its cross section per photon flux, with k and k' exchanged in the prefactor, $\sigma_-(k' \to k) = (32\pi^2e^2m^2\omega/3\hbar^3c)\,S'/(k'^3k)$. Here S' has the radial integrals $R'_{l\to l'} = \int u_{kl'}\,r\,u_{k'l}\,\mathrm{d}r = R_{l'\to l}$. Shifting the summation index in each of the two sums of Eq. (3.9) shows $S' = S$. Hence

$$\boxed{k^2\,\sigma_+(k \to k') = k'^2\,\sigma_-(k' \to k).} \tag{4.1}$$

This is the principle of detailed balance for free–free transitions.

4.2 The absorption coefficient

(1) Let $g(E)\mathrm{d}E$ be the fraction of electrons with energy in $\mathrm{d}E$, for an isotropic distribution normalized to $\int g\,\mathrm{d}E = 1$, and let $\sigma_-(E)$ denote the emission cross section from E to $E-\hbar\omega$, zero for $E < \hbar\omega$. The wave loses photons at the rate $n_e N\Phi \int g\,v(\sigma_+ - \sigma_-)\mathrm{d}E$ per unit volume, so its intensity decays with the coefficient

$$\boxed{\kappa = n_e N \int_0^\infty g(E)\,v(E)\big[\sigma_+(E) - \sigma_-(E)\big]\mathrm{d}E.} \tag{4.2}$$

(2) Shift the emission integral to $E = E'' + \hbar\omega$ and use Eq. (4.1), $\sigma_-(E''+\hbar\omega) = [E''/(E''+\hbar\omega)]\,\sigma_+(E'')$. For the Maxwell distribution $g(E) = 2\pi^{-1/2}(k_BT)^{-3/2}E^{1/2}e^{-E/k_BT}$,

$$g(E''+\hbar\omega)\,v(E''+\hbar\omega)\,\frac{E''}{E''+\hbar\omega} = e^{-u}\,g(E'')\,v(E''), \tag{4.3}$$

since $g\,v$ is proportional to $E\,e^{-E/k_BT}$. Therefore

$$\boxed{\kappa = n_e N\,(1-e^{-u})\int_0^\infty g(E)\,v(E)\,\sigma_+(E)\,\mathrm{d}E \qquad \text{(Maxwell electrons).}} \tag{4.4}$$

Stimulated emission reduces the absorption by the factor $1-e^{-u}$, as it does for ions [3].

4.3 The kinetic equation and its Fokker–Planck limit

(1) Let $n(E)\mathrm{d}E$ be the number of electrons per unit volume in $\mathrm{d}E$, and write $n = n_e E^{1/2}F(E)$ with $\int E^{1/2}F\,\mathrm{d}E = 1$, the normalization of the two-term Boltzmann codes [28]. An electron at E climbs to $E+\hbar\omega$ at the rate $W_+(E) = N\Phi v\sigma_+(E)$ and falls to $E-\hbar\omega$ at the rate $W_-(E) = N\Phi v\sigma_-(E)$. The master equation for the distribution is

$$\frac{\partial n(E)}{\partial t} = W_+(E-\hbar\omega)n(E-\hbar\omega) + W_-(E+\hbar\omega)n(E+\hbar\omega) - \big[W_+(E)+W_-(E)\big]n(E). \tag{4.5}$$

(2) Detailed balance, Eq. (4.1), makes the two rates that connect E and $E+\hbar\omega$ equal after weighting with $E^{1/2}$,

$$W_+(E)\,E^{1/2} = W_-(E+\hbar\omega)\,(E+\hbar\omega)^{1/2} = N\Phi\Big(\frac{2}{m}\Big)^{1/2}E\,\sigma_+(E) \equiv \mathcal{Q}\Big(E+\frac{\hbar\omega}{2}\Big). \tag{4.6}$$

Dividing Eq. (4.5) by n_e gives the jump operator in conservation form,

$$\boxed{\frac{\partial(E^{1/2}F)}{\partial t} = \mathcal{Q}\Big(E+\frac{\hbar\omega}{2}\Big)\big[F(E+\hbar\omega)-F(E)\big] - \mathcal{Q}\Big(E-\frac{\hbar\omega}{2}\Big)\big[F(E)-F(E-\hbar\omega)\big].} \tag{4.7}$$

The right side is a difference of two fluxes through the energies $E\pm\hbar\omega/2$, so the operator conserves the number of electrons.

(3) When $\hbar\omega$ is small against the scale on which F and $\mathcal{Q}$ vary, each bracket is $\hbar\omega\,\partial F/\partial E$ at the midpoint, and the difference of the two fluxes is $\hbar\omega$ times their derivative,

$$\frac{\partial(E^{1/2}F)}{\partial t} = \frac{\partial}{\partial E}\Big[\mathcal{D}(E)\,\frac{\partial F}{\partial E}\Big] + O\big((\hbar\omega)^4\big), \qquad \mathcal{D} = (\hbar\omega)^2\mathcal{Q}(E). \tag{4.8}$$

With the soft-photon cross section of Sec. 5, $\sigma_+ = 4\pi e^2 v^2 \sigma_{\rm m}/3c\hbar\omega^3$, and $\Phi = cE_0^2/8\pi\hbar\omega$,

$$\boxed{\mathcal{D}(E) = \frac{2}{3}\,\frac{e^2 E_{\rm rms}^2}{m\omega^2}\,N\Big(\frac{2}{m}\Big)^{1/2} E^2\,\sigma_{\rm m}(E), \qquad E_{\rm rms}^2 = \frac{E_0^2}{2}.} \tag{4.9}$$

This is the field term of the two-term Boltzmann equation in a high-frequency field [28] in the limit $\omega \gg \nu$. There the field diffusion is $\propto E_{\rm rms}^2\,\epsilon\,\sigma/[\sigma^2 + (\omega/N)^2/\gamma^2\epsilon]$ and tends to $E_{\rm rms}^2\epsilon^2\sigma/\omega^2$, the dependence $E^2\sigma_{\rm m}$ of Eq. (4.9). The check script confirms that both give the same heating rate. Equation (4.7) is therefore the form that a two-term code should use when $\hbar\omega$ is not small against the energies of the electrons.

5 The soft-photon limit

5.1 The force identity

(1) As $\omega \to 0$ the final state approaches the initial one, and Eq. (3.13) needs the integral of V' between two states of the same energy and neighbouring l. It is fixed by the phase shifts alone. Write $u_l = u_{kl}$, $u_{l+1} = u_{k,l+1}$ and $A_l = l(l+1)/r^2$, so that $u_l'' = (A_l + U - k^2)u_l$ by Eq. (3.2), and define

$$P = u_{l+1}u_l, \qquad h = u_{l+1}'u_l', \qquad g = u_{l+1}'u_l - u_{l+1}u_l'. \tag{5.1}$$

(2) The radial equations give $h' = (U-k^2)P' + A_{l+1}u_{l+1}u_l' + A_l u_{l+1}'u_l$. With $(U-k^2)P' = [(U-k^2)P]' - U'P$ this is

$$U'P = \big[(U-k^2)P\big]' - h' + A_{l+1}u_{l+1}u_l' + A_l u_{l+1}'u_l. \tag{5.2}$$

(3) The two centrifugal terms are $\frac{l+1}{r^2}\big[(l+2)u_{l+1}u_l' + l\,u_{l+1}'u_l\big] = \frac{l+1}{r^2}\big[(l+1)P' - g\big]$. The equations for u_l and u_{l+1} also give $g' = (A_{l+1} - A_l)P = 2(l+1)P/r^2$. Hence

$$\frac{(l+1)^2P'}{r^2} = \Big[\frac{(l+1)^2P}{r^2}\Big]' + \frac{2(l+1)^2P}{r^3} = \Big[\frac{(l+1)^2P}{r^2}\Big]' + \frac{(l+1)g'}{r}, \tag{5.3}$$

and $(l+1)g'/r - (l+1)g/r^2 = (l+1)(g/r)'$. Every term is a derivative,

$$U'(r)\,u_{l+1}u_l = \frac{\mathrm{d}}{\mathrm{d}r}\Big[(U-k^2)P - h + \frac{(l+1)^2P}{r^2} + \frac{(l+1)g}{r}\Big]. \tag{5.4}$$

(4) At $r = 0$ the regular solutions behave as r^{l+1} and r^{l+2}, so $P \sim r^{2l+3}$, h, P/r^2 and g/r all $\sim r^{2l+1}$, and the bracket vanishes for a potential less singular than r^{-3}. Where U has fallen off, $u_l = \sin\varphi$ and $u_{l+1} = -\cos(\varphi+\Delta)$ with $\varphi = kr - l\pi/2 + \delta_l$ and $\Delta = \delta_{l+1} - \delta_l$. Then $g \to k\cos\Delta$ is bounded, the terms with $1/r$ and $1/r^2$ vanish, and

$$-k^2P - h = k^2\big[\sin\varphi\cos(\varphi+\Delta) - \cos\varphi\sin(\varphi+\Delta)\big] = -k^2\sin\Delta, \tag{5.5}$$

a constant. Integrating Eq. (5.4) from 0 to ∞ and returning to $V = \hbar^2U/2m$,

$$\boxed{\int_0^\infty u_{k,l+1}\,V'(r)\,u_{kl}\,\mathrm{d}r = \frac{\hbar^2k^2}{2m}\,\sin(\delta_l - \delta_{l+1}).} \tag{5.6}$$

The check script verifies it for the deep models of argon and krypton of Sec. 8 at $ka_0 \geq 0.05$, to 10^{-4} for each $l \leq 2$ and to 4×10^{-6} for the sum over $l \leq 9$ weighted with $l+1$, as the integrals enter the cross sections below. The higher partial waves at small k carry integrals of 10^{-7} and less, which the integration resolves only to a few per cent.

5.2 The soft-photon cross section

(1) Insert Eq. (5.6) into Eq. (3.13) at $k' = k$. The terms with $l \to l+1$ and $l \to l-1$ of Eq. (3.9) give the same sum after a shift of the index,

$$S \to \left(\frac{\hbar^2 k^2}{2m^2\omega^2}\right)^2 2\sum_l (l+1)\sin^2(\delta_l - \delta_{l+1}) = \frac{\hbar^4 k^6}{8\pi m^4 \omega^4}\,\sigma_{\rm m}(E), \tag{5.7}$$

with the momentum-transfer cross section of Appendix A,

$$\sigma_{\rm m}(E) = \int (1-\cos\theta)\frac{{\rm d}\sigma}{{\rm d}\Omega}{\rm d}\Omega = \frac{4\pi}{k^2}\sum_{l=0}^{\infty}(l+1)\sin^2(\delta_l - \delta_{l+1}). \tag{5.8}$$

(2) Equation (3.11) becomes

$$\boxed{\sigma_+ \to \sigma_+^{\rm soft} = \frac{4\pi}{3}\,\frac{e^2 v^2}{c\hbar\omega^3}\,\sigma_{\rm m}(E) \qquad (\hbar\omega \ll E).} \tag{5.9}$$

This is the content of the low-energy theorem of Low [7] for potential scattering. Kroll and Watson [9] wrote it, angle by angle, as $(k'/k)(\mathbf{Q}\cdot\mathbf{a})^2/4$ times the elastic cross section, where $\mathbf{Q} = \mathbf{k} - \mathbf{k}'$ and $\mathbf{a} = eE_0\hat{\mathbf{e}}/m\omega^2$. The average of $(\mathbf{Q}\cdot\hat{\mathbf{e}})^2$ over directions is $Q^2/3$, and $Q^2 = 2k^2(1-\cos\theta)$ turns the elastic cross section into $\sigma_{\rm m}$, which is Eq. (5.9).
(3) The literature on neutral bremsstrahlung, the emission that corresponds to this absorption, is divided on this point. Buzulutskov et al. [16] report two opinions, the integral elastic cross section, used by Kas'yanov and Starostin [6] among others, and the momentum-transfer cross section, used by Dalgarno and Lane [10] among others, and they adopt the elastic one. The force identity decides for $\sigma_{\rm m}$. The two differ where the scattering is anisotropic, most of all in the Ramsauer windows. At the minimum of the deep models of Sec. 8 the elastic cross section is 3.8 times $\sigma_{\rm m}$ in argon and 6.6 times in krypton.

5.3 The classical limit is the Lorentz gas

(1) The classical kernel is the conductivity of the Lorentz gas. Expand the distribution of velocities as $f_0(v) + (\mathbf{v}\cdot\hat{\mathbf{e}}/v)f_1(v,t)$. The force on an electron is $-e\mathbf{E}$, and heavy scatterers relax the anisotropic part at the rate $\nu(v) = N\sigma_{\rm m} v$. The Boltzmann equation gives for $f_1 = {\rm Re}[\hat{f}_1 e^{-i\omega t}]$

$$(\nu - i\omega)\hat{f}_1 = \frac{eE_0}{m}\frac{\partial f_0}{\partial v}. \tag{5.10}$$

(2) The current along $\hat{\mathbf{e}}$ is $-e\int v\cos^2\theta\, f_1\,{\rm d}^3v = -(4\pi e/3)\int v^3 f_1\,{\rm d}v$, and with $\int 4\pi v^2 f_0\,{\rm d}v = n_e$ the conductivity is

$$\sigma(\omega) = -\frac{4\pi e^2}{3m}\int_0^\infty \frac{v^3}{\nu(v) - i\omega}\frac{\partial f_0}{\partial v}\,{\rm d}v. \tag{5.11}$$

For a constant ν an integration by parts returns the Drude conductivity $e^2 n_e/m(\nu - i\omega)$ [1].
(3) For $\omega \gg \nu$, ${\rm Re}(\nu - i\omega)^{-1} \to \nu/\omega^2$. For Maxwell electrons $\partial f_0/\partial v = -(mv/k_B T)f_0$, and the absorption coefficient $\kappa = 4\pi\,{\rm Re}\,\sigma/c$ at refractive index one is

$$\boxed{\kappa_{\rm cl} = \frac{4\pi e^2 n_e N}{3c\,k_B T\,\omega^2}\,\langle v^3\sigma_{\rm m}\rangle, \qquad \langle X\rangle = \int_0^\infty g(E)\,X\,{\rm d}E.} \tag{5.12}$$

(4) The quantum result reduces to it. In Eq. (4.4) put the soft-photon cross section, Eq. (5.9), and $1-e^{-u}\to u$,

$$\kappa \to n_e N\,\frac{\hbar\omega}{k_B T}\,\frac{4\pi e^2}{3c\hbar\omega^3}\,\langle v^3\sigma_{\rm m}\rangle = \kappa_{\rm cl}. \tag{5.13}$$

The check script confirms the equality numerically with the measured cross sections.

5.4 The free–free Gaunt factor and the error of the classical kernel

(1) Define the free–free Gaunt factor of the electron–atom pair by $\sigma_+ = G(E,\omega)\,\sigma_+^{\rm soft}$. With Eqs. (4.4) and (5.12) the ratio of the quantum to the classical absorption coefficient is

$$\boxed{\mathcal{R}(T,\omega) = \frac{\kappa}{\kappa_{\rm cl}} = \frac{1-e^{-u}}{u}\,\frac{\langle v^3\sigma_{\rm m} G\rangle}{\langle v^3\sigma_{\rm m}\rangle}.} \tag{5.14}$$

The average is over $g\,v^3 \propto E^2 e^{-E/k_BT}$. The quantity $\mathcal{R}-1$ is the error of the classical kernel that Sec. 1 asks for. Two effects compete. The factor $(1-e^{-u})/u$ is below one, and G exceeds one for slow electrons because a photon larger than their energy carries them to where $\sigma_{\rm m}$ is different.

6 Exact limits at finite photon energy

6.1 The zero-range potential

(1) A potential of zero range scatters only the s wave, with $k\cot\delta_0 = -1/a$ for the scattering length a. Outside the origin $u_{k0} = \sin(kr+\delta_0)$ and $u_{kl} = \hat\jmath_l(kr)$ for $l \ge 1$. The only transitions are $0\to 1$ and $1\to 0$.
(2) Write $\sin(kr+\delta_0) = \cos\delta_0 \sin kr + \sin\delta_0\cos kr$. The part with $\sin kr = \hat\jmath_0(kr)$ is a free–free integral between plane waves and vanishes for $k \neq k'$. Appendix B evaluates the other part with the convergence factor,

$$\int_0^\infty \hat\jmath_1(k'r)\,r\cos(kr)\,{\rm d}r = \frac{2k'^2}{(k'^2-k^2)^2}, \tag{6.1}$$

so that, with $k'^2 - k^2 = 2m\omega/\hbar$,

$$R_{0\to1} = \frac{\hbar^2 k'^2}{2m^2\omega^2}\sin\delta_0(k), \qquad R_{1\to0} = \frac{\hbar^2 k^2}{2m^2\omega^2}\sin\delta_0(k'). \tag{6.2}$$

(3) Then $S = R_{0\to1}^2 + R_{1\to0}^2$ and, with $\sigma_{\rm m} = (4\pi/k^2)\sin^2\delta_0$ from Eq. (5.8), Eq. (3.11) gives

$$\boxed{\sigma_+^{\rm zr} = \frac{2\pi}{3}\,\frac{e^2\hbar\,kk'}{c\,m^2\omega^3}\Big[\Big(\frac{k'}{k}\Big)^2\sigma_{\rm m}(E) + \sigma_{\rm m}(E')\Big], \qquad E' = E+\hbar\omega.} \tag{6.3}$$

The cross section at both energies enters, the initial one weighted by $(k'/k)^2$. As $\omega\to 0$ the bracket tends to $2\sigma_{\rm m}(E)$ and Eq. (5.9) returns. The check script compares Eq. (6.2) with the acceleration form for a Gaussian well of width $b\to 0$ tuned to $a = -1.46\,a_0$ and finds the deviation falling in proportion to b, from 1.1×10^{-2} at $b = 0.1\,a_0$ to 2.2×10^{-3} at $b = 0.025\,a_0$.
(4) Written for emission with Eq. (4.1) and multiplied by the density $8\pi\nu^2/c^2$ of photon modes per unit frequency, which turns stimulated into spontaneous emission, Eq. (6.3) becomes

$$\frac{{\rm d}\sigma}{{\rm d}\nu} = \frac{8}{3}\,\frac{r_e}{c}\,\frac{1}{h\nu}\Big(\frac{E-h\nu}{E}\Big)^{1/2}\big[(E-h\nu)\,\sigma_{\rm m}(E) + E\,\sigma_{\rm m}(E-h\nu)\big], \qquad r_e = \frac{e^2}{mc^2}, \tag{6.4}$$

for an electron of energy E that emits a photon of frequency ν. This is Eq. (4) [16], the cross section for neutral bremsstrahlung of Buzulutskov et al., with $\sigma_{\rm m}$ in place of their elastic cross section. They attribute it to the Born approximation with one dominant partial wave. Equation (6.3) shows that it is exact for a zero-range potential of any strength, for which the elastic and momentum-transfer cross sections coincide. The check script confirms the prefactors to 10^{-13}.

6.2 The Born approximation

(1) In the first Born approximation the one-photon cross section of Bunkin and Fedorov [4] holds for any potential, with the Born amplitude $f_B(Q) = -(m/2\pi\hbar^2)\tilde{V}(Q)$. Dividing by Φ and averaging $(\mathbf{Q}\cdot\hat{\mathbf{e}})^2$ over directions,

$$\sigma_+^B = \frac{2\pi\hbar e^2}{3cm^2\omega^3}\,\frac{k'}{k}\int Q^2 |f_B(Q)|^2\,\mathrm{d}\Omega'. \tag{6.5}$$

(2) At fixed k and k', $Q^2 = k^2 + k'^2 - 2kk'\cos\theta$ gives $\mathrm{d}\Omega' = 2\pi Q\,\mathrm{d}Q/kk'$ with Q from $k'-k$ to $k'+k$. With $M(q) = \int_0^q Q^3|f_B|^2\mathrm{d}Q$,

$$\sigma_+^B = \frac{4\pi^2\hbar e^2}{3cm^2\omega^3k^2}\big[M(k'+k) - M(k'-k)\big]. \tag{6.6}$$

(3) The elastic momentum-transfer cross section at momentum $\hbar\kappa$ is, with $Q^2 = 2\kappa^2(1-\cos\theta)$ and $\mathrm{d}\Omega = 2\pi Q\,\mathrm{d}Q/\kappa^2$, $\sigma_{\rm m}^B(\kappa) = (\pi/\kappa^4)M(2\kappa)$. Hence

$$\boxed{\sigma_+^B = \frac{4\pi\hbar e^2}{3cm^2\omega^3k^2}\Big[\kappa_+^4\,\sigma_{\rm m}^B(\kappa_+) - \kappa_-^4\,\sigma_{\rm m}^B(\kappa_-)\Big], \qquad \kappa_\pm = \frac{k'\pm k}{2}.} \tag{6.7}$$

In the Born approximation the exact one-photon cross section at any photon energy is the momentum-transfer cross section at the mean momentum $(k+k')/2$, less a term at the half difference. The check script confirms Eq. (6.7) to 10^{-14}. Section 8 finds that the same form with the true $\sigma_{\rm m}$ is the best of the formulas that use only $\sigma_{\rm m}$, far outside the Born approximation.

6.3 The first order in the photon energy

(1) Expand Eqs. (6.3) and (6.7) to first order in $\hbar\omega/E$. With $\bar{E} = E + \hbar\omega/2$ the mean energy, $\kappa_+^2 = k^2 + m\omega/\hbar + O(\omega^2)$, so $\hbar^2\kappa_+^2/2m = \bar{E}$ to first order and κ_-^4 is of fourth order. In Eq. (6.3), $\sigma_{\rm m}(E)$ and $\sigma_{\rm m}(E')$ are $\sigma_{\rm m}(\bar{E}) \mp \frac{1}{2}\hbar\omega\sigma'_{\rm m}$, and the terms with $\sigma'_{\rm m}$ cancel against $k^2 - k'^2$ to second order. Both give

$$\boxed{\sigma_+ = \frac{4\pi}{3}\,\frac{e^2\hbar\,k'^2}{c\,m^2\omega^3}\,\sigma_{\rm m}(\bar{E})\Big[1 + O\Big(\frac{\hbar\omega}{E}\Big)^2\Big].} \tag{6.8}$$

(2) The two exact limits agree to first order. The theorem of Burnett and Kroll [8] states that the term of first order beyond the soft-photon limit is fixed by the elastic cross section alone. For the model potentials of Sec. 8, which are neither weak nor of zero range, the check script finds that the residual of Eq. (6.8) falls by a factor between 3.85 and 4.04 each time $\hbar\omega/E$ is halved, the signature of a second-order remainder.

7 Strong fields and the weak-field criterion

(1) Kroll and Watson [9] extended the soft-photon result to any field strength. For a collision short against the period of the wave and $\ell\hbar\omega \ll E$, the cross section for the exchange of ℓ photons is

$$\frac{\mathrm{d}\sigma_\ell}{\mathrm{d}\Omega'} = \frac{k_\ell}{k}\,J_\ell^2(\mathbf{Q}\cdot\mathbf{a})\,\frac{\mathrm{d}\sigma_{\rm el}}{\mathrm{d}\Omega}, \tag{7.1}$$

with the elastic cross section at shifted momenta. It has the structure of the Born result of Bunkin and Fedorov [4], with the Born cross section replaced by the exact one. Weingartshofer et al. observed the multiphoton terms directly [17].
(2) One photon dominates when $|\mathbf{Q}\cdot\mathbf{a}| \ll 1$, with $a = eE_0/m\omega^2$ and $E_0 = (8\pi I/c)^{1/2}$. Since $Q \le 2k$, the condition $2ka \le 0.3$ keeps J_1^2 within 2.3 % of $x^2/4$. Table 2 gives the intensity at which it is reached for an electron of energy $\hbar\omega$, for which the quantum corrections of Secs. 8 and 9 matter most, and for electrons of energy $3k_BT/2$.

Table 2: Intensity in $\mathrm{W\,cm^{-2}}$ at which $2ka = 0.3$, the edge of the one-photon regime, for an electron of energy $\hbar\omega$ and of energy $3k_BT/2$ at 300 K and at 0.63 eV.

frequency	$E = \hbar\omega$	$E = 3k_BT/2$, 300 K	$E = 3k_BT/2$, 0.63 eV
1 THz	1.4×10^3	1.5×10^2	6.1×10^0
3 THz	3.7×10^4	1.2×10^4	4.9×10^2
10 THz	1.4×10^6	1.5×10^6	6.1×10^4
28.3 THz	3.1×10^7	9.5×10^7	3.9×10^6

(3) Above that intensity the multiphoton terms of Eq. (7.1) appear. For electrons with $\hbar\omega \ll E$ they change how the energy is exchanged but not how fast it is gained. The energy diffusion is proportional to $\sum_\ell \ell^2 J_\ell^2(x)$, and the sum rule $\sum_\ell \ell^2 J_\ell^2(x) = x^2/2$, which follows from the Jacobi–Anger expansion and Parseval's theorem, makes it equal to the one-photon value $2 \cdot x^2/4$. The classical kernel therefore holds at any intensity for those electrons. With the field above the collision frequency and only elastic losses the electrons are Maxwellian, and their temperature exceeds that of the gas, T_g, by $k_B(T_e - T_g) = \frac{2}{3}(M/m)U_p$, with the quiver energy $U_p = e^2E_0^2/4m\omega^2$ [5]. A jump of 0.5 eV in pure krypton then needs 53 $\mathrm{W\,cm^{-2}}$ at 0.3 THz and 585 $\mathrm{W\,cm^{-2}}$ at 1 THz, where $2ka$ is 7.1 and 2.1 for an electron of 0.5 eV, and it heats as the classical theory says. The one-photon corrections of these notes need $2ka \le 0.3$ only for the slow electrons, which holds up to 1.4 $\mathrm{kW\,cm^{-2}}$ at 1 THz and 31 $\mathrm{MW\,cm^{-2}}$ at 28.3 THz.

8 Realistic atoms

8.1 Three families of model potentials

(1) The exact cross section, Eq. (3.11), needs the radial functions of a potential that reproduces the measured scattering. Three local forms are fitted, in atomic units,

$$\begin{aligned}
&\text{A:}\quad V = V_0 e^{-r^2/b^2} - \frac{\alpha_d}{2(r^2+r_c^2)^2},\\
&\text{B:}\quad V = V_0 e^{-r/b} - \frac{\alpha_d\,(1-e^{-(r/r_c)^6})}{2r^4},\\
&\text{D:}\quad V = -\frac{Z}{r}\,\frac{1}{1+H(e^{r/d}-1)} - \frac{\alpha_d}{2(r^2+r_c^2)^2}.
\end{aligned} \tag{8.1}$$

A and B have a repulsive core. They bind no states, except that krypton's model B, whose cut-off $r_c = 0.36\,a_0$ leaves the polarization attraction strong inside the atom, binds one s and one p state. D has the screening function of the independent-particle model of Green, Sellin and Zachor [27] for the nucleus of charge Z, and is constrained to bind the occupied orbitals of the atom, two s and

one p state for neon, three s and two p for argon, four s, three p and one d for krypton. The count is the number of nodes of the zero-energy solution. All three have the polarization tail of Eq. (2.1) with the recommended static polarizabilities, 2.6611, 11.083 and 16.78 atomic units for Ne, Ar and Kr [22].

(2) The parameters minimize $\sum(\ln\sigma_{\mathrm{m}}^{\mathrm{model}} - \ln\sigma_{\mathrm{m}}^{\mathrm{LXCat}})^2$ over 36 energies from 0.003 to 4 eV, against the measured cross sections that the classical kernel uses, the BSR set for neon, the IST-Lisbon set for argon and the Biagi set for krypton, taken from LXCat [30]. Table 3 lists the quality of the fits and the scattering lengths, and Fig. 1 compares the cross sections. The deep models D fit within 7 to 11 % and reproduce the Ramsauer windows of argon and krypton and the rise toward their d-wave shape resonances. The repulsive cores cannot dig a window as deep as krypton's, and model A misses it by a factor of 1.75 on average. The scattering length of D lies inside the range of the measurements for krypton and $0.024\,a_0$ beyond its edge for argon [24, 25, 23]. For neon the three models give 0.28 to $0.29\,a_0$, between the measured $0.21\,a_0$ [26] and the $0.38\,a_0$ implied by the zero-energy cross section of the BSR set, which they are fitted to. The scattering lengths come from the zero-energy solution, whose polarization tail $-\alpha_d/2r^4$ is followed to $4000\,a_0$ and continued analytically beyond.

Table 3: The model potentials of Eq. (8.1). The rms of $\ln\sigma_{\mathrm{m}}^{\mathrm{model}} - \ln\sigma_{\mathrm{m}}^{\mathrm{LXCat}}$ over 0.003 to 4 eV, the scattering length A in a_0, and the range of measured scattering lengths quoted in [23] for argon and krypton and in [26, 23] for neon.

		rms of $\ln\sigma_{\mathrm{m}}$			A (a_0)			
gas	α_d	A	B	D	A	B	D	measured
Ne	2.6611	0.115	0.140	0.107	+0.285	+0.277	+0.288	+0.206 to +0.214
Ar	11.083	0.266	0.316	0.081	−1.450	−1.490	−1.483	−1.459 to −1.365
Kr	16.78	0.559	0.178	0.067	−3.087	−3.205	−3.254	−3.430 to −3.060

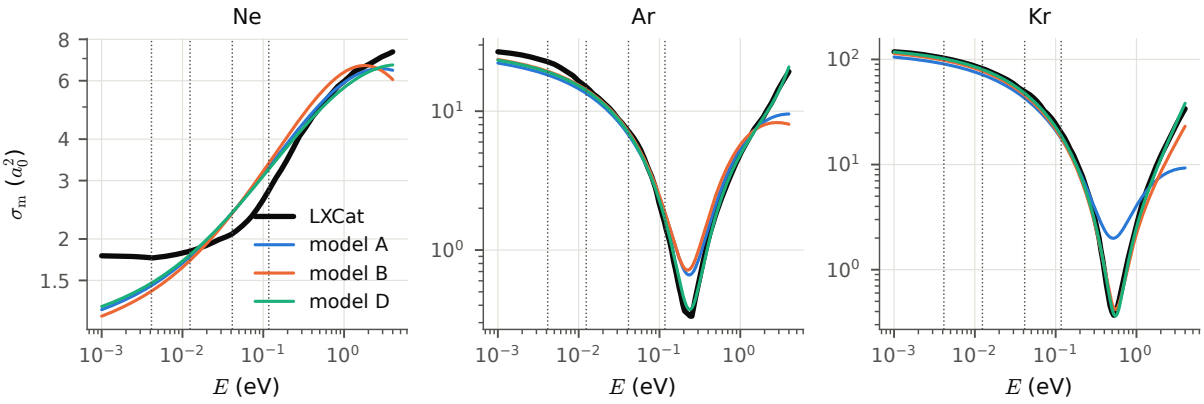


Figure 1: Momentum-transfer cross sections of the LXCat sets used by the classical kernel and of the three model families. The vertical lines mark the photon energies of 1, 3, 10 and 28.3 THz.

8.2 Numerical method

(1) The radial equation is integrated by the method of Numerov, on a uniform grid of step $0.01\,a_0$ for the smooth cores and on the logarithmic grid $r = e^x$, $u = e^{x/2}y$, with $y'' = [(l+\frac{1}{2})^2 + r^2(2V - k^2)]y$ and step $\Delta x = 5 \times 10^{-4}$, for the Coulomb-like cores of D. At $r_{\max} = 400\,a_0$ the solution is matched to the Riccati–Bessel functions, which fixes δ_l and the amplitude, and the acceleration integrals of Eq. (3.13) are accumulated during the integration and normalized at the end. Partial waves up to $l = 12$ are kept.
(2) For the deep models, halving the step, doubling $r_{\max}$ and raising the largest l to 16 change G by at most 1.4×10^{-5}. The cores of the models A reach 6×10^{10} hartree in krypton, where the recursion of Numerov is unstable on the uniform grid. The outward growth of the solution through the forbidden core filters that error out. Starting instead at a hard wall placed where $h^2 \cdot 2mV/\hbar^2$ last exceeds 2, with half the step, changes G of the models A and B by at most 1×10^{-5}. The force identity holds as stated in Sec. 5.1, and detailed balance, Eq. (4.1), to 10^{-13}.

8.3 The free–free Gaunt factor

(1) Figure 2 shows $G = \sigma_+/\sigma_+^{\text{soft}}$ for the deep models. For $\hbar\omega \ll E$ it tends to one, with the slope $1 + \frac{1}{2}\,\mathrm{d}\ln\sigma_{\mathrm{m}}/\mathrm{d}\ln E$ in $\hbar\omega/E$ that Eq. (6.8) predicts. Below $E \approx \hbar\omega$ it grows as $E^{-3/2}$. There the final state is near $\hbar\omega$ whatever the initial energy, σ_+ varies as $1/k$, and the soft-photon reference vanishes as k^2.

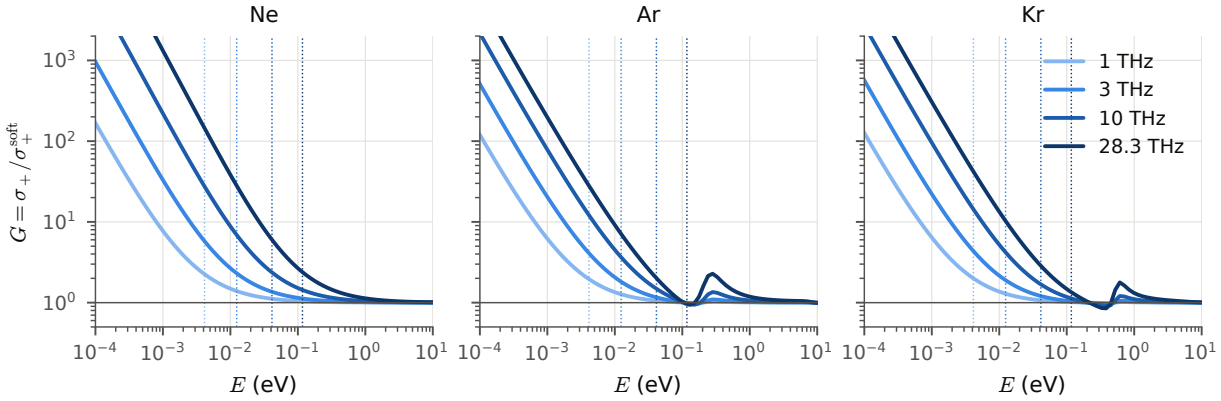


Figure 2: The free–free Gaunt factor $G = \sigma_+/\sigma_+^{\text{soft}}$ of the deep models at 1, 3, 10 and 28.3 THz. The dotted lines mark $E = \hbar\omega$.

(2) A code that needs σ_+ without solving the scattering problem can use one of three forms built on σ_{m} alone, the two-energy form of Eq. (6.3), the mean-energy form of Eq. (6.8), or the mean-momentum form of Eq. (6.7) with the true σ_{m}. Table 4 compares them with the exact result of each deep model. The two-energy form, exact for the zero-range potential, overestimates σ_+ of argon by a factor 2.1 for thermal electrons at 28.3 THz. The polarization tail makes the p wave matter at low energy, since it raises the p-wave phase shift as $\pi\alpha_d k^2/15$ in atomic units rather than as k^3 [21], a law the check script confirms for the deep models. The mean-energy form falls up to 25 % short there.
(3) The mean-momentum form is the best of the three. For neon it is within 0.5 % of the exact result at every energy above $\hbar\omega$ up to 28.3 THz. For argon and krypton it is within 1.2 % at 3 THz

and 3.3 % at 10 THz above $\hbar\omega$. At 28.3 THz it errs by up to 14 % in argon and 17 % in krypton above $\hbar\omega$, most where the final energy $E+\hbar\omega$ falls at the Ramsauer minimum of the model, 0.24 eV in argon and 0.55 eV in krypton. There $\sigma_{\rm m}$ changes by a large factor across one photon energy. Table 4 shows errors of the same size for thermal electrons below $\hbar\omega$. In the thermal averages of Sec. 9 up to 4 THz the form is within 1 % of the exact result for every gas.

Table 4: The exact Gaunt factor G of the deep models at $E = 26$ meV and 0.3 eV, and the two-energy (TE), mean-energy (ME) and mean-momentum (MM) forms with the model's own $\sigma_{\rm m}$, as ratios to the exact cross section.

			$E = 26$ meV			$E = 0.3$ eV	
gas	f (THz)	G	TE	ME	MM	G	MM
Ne	3	1.588	0.989	1.002	0.999	1.048	1.000
Ne	10	3.345	0.935	1.009	0.996	1.162	1.000
Ne	28.3	10.373	0.805	1.019	0.985	1.479	0.999
Ar	3	1.318	1.030	0.989	0.996	1.095	1.000
Ar	10	1.984	1.223	0.927	0.968	1.350	0.997
Ar	28.3	3.144	2.092	0.747	0.859	2.219	0.987
Kr	3	1.370	1.017	0.989	0.994	0.979	0.999
Kr	10	2.252	1.121	0.927	0.955	0.939	0.986
Kr	28.3	4.478	1.502	0.757	0.834	0.882	0.891

8.4 Polarization bremsstrahlung

(1) The term of Eq. (2.3) has the same angular structure as $\mathbf{r}$, so in the acceleration form it adds $-m\omega^2\alpha_d/r^2$ to $V'(r)$. With the cut-off $r^2 \to r^2 + r_c^2$ of the polarization potential of model D, Table 5 gives the change of σ_+. It is below 0.7 % up to 3 THz for every gas and energy. At 28.3 THz it reaches -13 % for slow electrons in argon, -8 % in krypton and $+4$ % in neon, so a CO_2-laser calculation for a cold gas has to include it. At 1 eV it stays below 1 % at every frequency. Geltman [20] studied the same field-induced dipole in the laser-assisted elastic scattering of Sec. 7.

9 Thermal absorption and the error of the classical kernel

9.1 The pure gases

(1) Equation (5.14) gives $\mathcal{R}(T,\omega)$ from G and $\sigma_{\rm m}$ of each model. Table 6 lists it for the deep models with the range over the three families. Up to 4 THz the classical kernel is right to 2.5 % or better for every gas and temperature, and to 2.8 % with any of the three families. At 10 THz electrons at 300 K absorb 12 % less than the classical kernel says in argon and 7 % less in krypton, and at 28.3 THz 41 % and 25 % less. In neon the two effects of Sec. 5 cancel to within 3 %. Above 0.3 eV the classical kernel is right to 1.6 % at every frequency up to 28.3 THz.

9.2 The two fills

(1) In a mixture each gas contributes to $\kappa_{\rm cl}$ in proportion to its share $x_g\langle v^3\sigma_{\rm m}\rangle$, and the ratio of the mixture is the share-weighted mean of the ratios of its gases. Fluorine appears only at 0.2 and

Table 5: Change of σ_+ in percent when the polarization bremsstrahlung of Eq. (2.3) is added, for the deep models.

gas	E (meV)	1 THz	3 THz	10 THz	28.3 THz
Ne	3	+0.09	+0.43	+1.66	+4.33
Ne	26	+0.01	+0.10	+0.72	+2.84
Ne	300	+0.00	+0.01	+0.06	+0.44
Ne	1000	+0.00	+0.00	+0.01	+0.10
Ar	3	−0.12	−0.63	−3.15	−12.82
Ar	26	−0.03	−0.26	−2.28	−12.68
Ar	300	−0.00	−0.02	−0.07	+0.93
Ar	1000	+0.00	+0.01	+0.09	+0.66
Kr	3	−0.08	−0.41	−2.01	−7.53
Kr	26	−0.02	−0.15	−1.29	−6.68
Kr	300	−0.01	−0.08	−0.95	−7.70
Kr	1000	+0.00	+0.01	+0.12	+0.93

0.3 %, and its molecular cross section is treated with the mean-momentum form. At 300 K krypton carries 44 % of the classical absorption of the neon fill and 83 % of that of the argon fill, because per atom its $\langle v^3 \sigma_{\mathrm{m}} \rangle$ is 15 times that of neon and 7.6 times that of argon.
(2) Table 7 gives the ratio for both fills, from the deep models and, in parentheses, from the mean-momentum form with the measured cross sections alone. Figure 3 shows it against frequency.

9.3 The answer for the laser gas

(1) In the band of 0.3 to 4 THz proposed for reheating, the classical kernel differs from the quantum result by at most 1.6 % in either fill at any electron temperature from 300 K to 1.08 eV. The mean-momentum form with the measured cross sections gives at most 2.5 %. Both are below the 5 % of the criterion of Sec. 1, so the classical kernel holds in that band, and the absorption lengths and reflectances computed with it stand.
(2) For a CO_2 laser on a thermalized fill the classical kernel fails. At 28.3 THz and 300 K the quantum absorption is 10 % below it in the neon fill and 28 % below it in the argon fill, before the polarization bremsstrahlung of Sec. 8.4. Above 0.3 eV it is right to 1 % in both fills at every frequency studied. A code that heats electrons at such frequencies should multiply the classical absorption by $\mathcal{R}$, or replace the diffusion term of the two-term equation by the jump operator of Eq. (4.7) with the mean-momentum cross section.

10 Electron–ion or electron–atom

(1) The ions contribute the effective collision frequency $\nu_T \Lambda$ [1], with

$$\nu_T = \frac{4\sqrt{2\pi}}{3} \frac{n_i Z^2 e^4}{m^{1/2} (k_B T)^{3/2}}, \qquad \Lambda = \frac{\pi}{\sqrt{3}} \langle g \rangle \frac{1 - e^{-u}}{u},$$

the thermal Gaunt factor $\langle g \rangle$ times the factor of stimulated emission [2, 3]. The atoms contribute κ of Eq. (4.4), which corresponds to the effective collision frequency $(mN/3k_BT)\langle v^3 \sigma_{\mathrm{m}} \rangle \mathcal{R}$. With

Table 6: The ratio $\mathcal{R} = \kappa/\kappa_{\rm cl}$ of Eq. (5.14) for each gas, from the deep model D, and its range over the models A, B and D.

		$T = 300$ K		$T = 0.1$ eV		$T = 0.63$ eV	
gas	f (THz)	D	A–D	D	A–D	D	A–D
Ne	1	1.000	1.000–1.000	1.000	1.000–1.000	1.000	1.000–1.000
Ne	3	0.999	0.999–0.999	1.000	1.000–1.000	1.000	1.000–1.000
Ne	4	0.998	0.998–0.999	1.000	1.000–1.000	1.000	1.000–1.000
Ne	10	0.992	0.992–0.995	1.000	1.000–1.000	1.000	1.000–1.000
Ne	28.3	1.030	1.030–1.043	0.997	0.997–0.998	1.000	1.000–1.000
Ar	1	0.998	0.998–0.998	1.000	1.000–1.000	1.000	1.000–1.000
Ar	3	0.985	0.984–0.985	0.999	0.998–0.999	1.000	1.000–1.000
Ar	4	0.975	0.972–0.975	0.998	0.997–0.998	1.000	1.000–1.000
Ar	10	0.881	0.868–0.881	0.989	0.983–0.989	1.000	1.000–1.000
Ar	28.3	0.591	0.540–0.591	0.933	0.903–0.933	1.001	1.001–1.001
Kr	1	0.999	0.999–0.999	1.000	1.000–1.000	1.000	1.000–1.000
Kr	3	0.992	0.990–0.992	1.000	0.999–1.000	1.000	1.000–1.000
Kr	4	0.985	0.983–0.985	0.999	0.998–0.999	1.000	1.000–1.000
Kr	10	0.928	0.915–0.928	0.994	0.984–0.994	1.001	1.000–1.001
Kr	28.3	0.749	0.697–0.749	0.943	0.887–0.943	1.004	1.000–1.004

singly charged ions, $n_i = n_e$, the ions win above the electron density at which the two are equal.
(2) In the Born approximation the thermal Gaunt factor is $\langle g\rangle = (\sqrt{3}/\pi)e^{u/2}K_0(u/2)$ [3], valid for $Ze^2/\hbar v \ll 1$. For an electron of energy k_BT, $Ze^2/\hbar v$ is 23 at 300 K and 3.5 at 1.08 eV, and the collisions are classical. For $x = \omega Ze^2/mv^3 \ll 1$ and $\hbar\omega \ll k_BT$ the classical thermal Gaunt factor is that of Scheuer [29], $\langle g\rangle = (\sqrt{3}/\pi)\{\ln[(2k_BT)^{3/2}/\pi Ze^2m^{1/2}f] - \frac{5}{2}\gamma\}$, with f in hertz. At 300 K and 1 THz, x is about 5 for $v = (k_BT/m)^{1/2}$, and that formula turns negative. For larger x the classical Gaunt factor tends to Kramers' value one from above. The estimate used here is the larger of Scheuer's value and one. At 1 THz, where $u \le 0.16$, the Born value exceeds this estimate at every temperature and bounds the ionic absorption from above. At 28.3 THz and 300 K, where $u = 4.5$, the Born value falls to 0.44, below Kramers' one, which is then the better estimate.
(3) Table 8 and Fig. 4 give the density at which the two kinds of absorption are equal at 1 THz. Ions dominate only in a cold and dense plasma. At 300 K that takes 7.0×10^{19} m^{-3} in the neon fill and 1.5×10^{20} m^{-3} in the argon fill with the classical Gaunt factor, and at 0.63 eV more than 10^{22} m^{-3} in both. At 28.3 THz and 300 K, with Kramers' one, the densities are 2.7×10^{20} and 4.5×10^{20} m^{-3}. The plasma of the laser gas with the beam on, 2.7×10^{21} m^{-3} at 1.08 eV, absorbs on atoms, and so does the afterglow while its electrons stay warm. In a thermalized afterglow at 10^{20} m^{-3} and 1 THz the ions absorb 1.4 times as much as the atoms in the neon fill and 0.7 times as much in the argon fill. The atoms dominate because n_e/N is small only away from that corner.

11 Summary

(1) The one-photon free–free cross section of an electron in the field of an atom is Eq. (3.11), a sum over partial waves of the squared dipole integrals, which the acceleration form, Eq. (3.13),

Table 7: The ratio $\mathcal{R} = \kappa/\kappa_{\mathrm{cl}}$ for the two fills of Sec. 1, from the deep models, and in parentheses from the mean-momentum form with the LXCat cross sections.

fill	f (THz)	$T = 300$ K	$T = 0.1$ eV	$T = 0.63$ eV	$T = 1.08$ eV
Ne	0.3	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ne	1	1.000 (0.999)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ne	3	0.995 (0.992)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ne	4	0.992 (0.987)	1.000 (0.999)	1.000 (1.000)	1.000 (1.000)
Ne	5	0.988 (0.981)	1.000 (0.999)	1.000 (1.000)	1.000 (1.000)
Ne	10	0.961 (0.941)	0.999 (0.996)	1.000 (1.000)	1.000 (1.000)
Ne	28.3	0.897 (0.835)	0.991 (0.979)	1.000 (1.000)	1.000 (1.000)
Ar	0.3	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ar	1	0.999 (0.998)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Ar	3	0.991 (0.985)	1.000 (0.997)	1.000 (1.000)	1.000 (1.000)
Ar	4	0.984 (0.975)	0.999 (0.995)	1.000 (1.000)	1.000 (1.000)
Ar	5	0.975 (0.963)	0.998 (0.993)	1.000 (1.000)	1.000 (1.000)
Ar	10	0.920 (0.886)	0.993 (0.973)	1.000 (1.000)	1.000 (1.000)
Ar	28.3	0.722 (0.614)	0.941 (0.842)	1.003 (1.000)	1.001 (1.000)

makes absolutely convergent. Emission follows from detailed balance, Eq. (4.1), and the absorption coefficient of Maxwell electrons carries the factor $1 - e^{-u}$, Eq. (4.4). The kinetic equation is the jump operator of Eq. (4.7), whose Fokker–Planck limit, Eq. (4.9), is the field term of the two-term Boltzmann equation at $\omega \gg \nu$.

(2) The force identity, Eq. (5.6), reduces the soft-photon limit to the momentum-transfer cross section, Eq. (5.9), and the classical limit is the Lorentz gas, Eq. (5.12). The cross section that belongs in the classical formula is therefore σ_{m}, which settles the choice between σ_{m} and the elastic cross section that the literature on neutral bremsstrahlung leaves open [16].

(3) At finite photon energy the cross section is known exactly for a zero-range potential, Eq. (6.3), which is also the emission formula of neutral bremsstrahlung, Eq. (6.4), and in Born approximation, Eq. (6.7). Both agree to first order with Eq. (6.8), and the model potentials confirm that this order is universal. Wherever $\hbar\omega \leq E$ the mean-momentum form with the true σ_{m} is accurate to 0.5 % for neon at every frequency and to 3.3 % for argon and krypton up to 10 THz, but errs by up to 17 % at 28.3 THz near the Ramsauer minima.

(4) For the laser gas the quantum correction to the classical kernel is below 1.6 % from 0.3 to 4 THz at every temperature, so the classical kernel holds in that band. It becomes -10 to -28 % at 28.3 THz for electrons at 300 K, where polarization bremsstrahlung changes the cross section of slow electrons by up to -13 %. Electron–ion absorption competes only in a cold afterglow above about 10^{20} m^{-3}.

(5) Left open are the exchange of the scattering electron with the atomic electrons, which a one-body potential only mimics and which the multichannel R-matrix calculation of Bell and Berrington [13] treats for argon, the molecular fluorine, and the laser-assisted measurements on argon that find photon exchange, up to 12 % of the field-free cross section, where the theory of Kroll and Watson predicts none [18, 19]. The measured neutral bremsstrahlung [14, 15] is the experimental test of the models at higher photon energies.

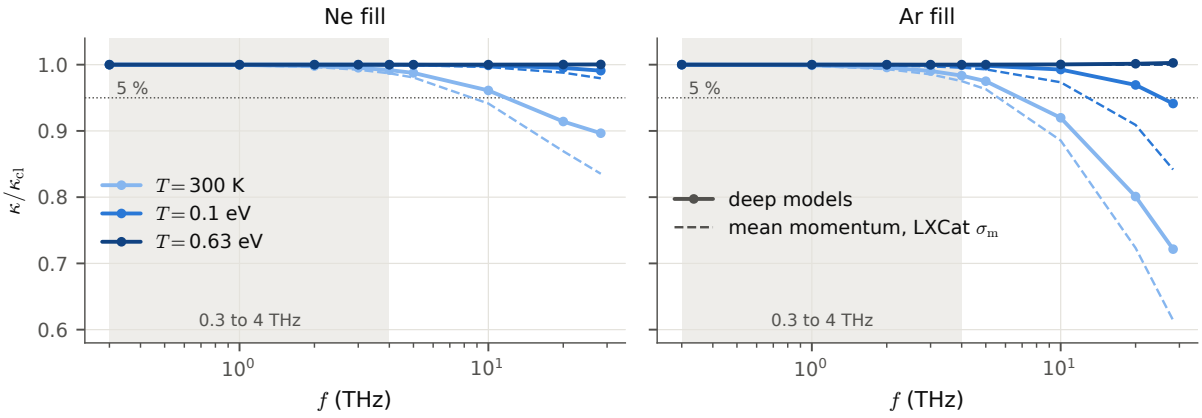


Figure 3: The ratio of the quantum to the classical absorption coefficient for the two fills against frequency. Solid lines are the deep models, dashed lines the mean-momentum form with the LXCat cross sections. The shaded band is the band of 0.3 to 4 THz proposed for reheating.

A Partial waves

(1) Take $\hat{\mathbf{e}} = \hat{z}$, which the average over the direction of the electron allows. The matrix elements of $\cos\theta$ are $\langle l+1, m|\cos\theta|lm\rangle = [((l+1)^2 - m^2)/((2l+1)(2l+3))]^{1/2}$ and vanish for $m' \neq m$. Summing over m,

$$\sum_{m=-l}^{l} \frac{(l+1)^2 - m^2}{(2l+1)(2l+3)} = \frac{(2l+1)(l+1)^2 - l(l+1)(2l+1)/3}{(2l+1)(2l+3)} = \frac{l+1}{3}, \tag{A.1}$$

and the same steps with $\langle l-1, m|\cos\theta|lm\rangle^2 = (l^2-m^2)/((2l-1)(2l+1))$ give $l/3$, which is Eq. (3.6). (2) The elastic amplitude is $f(\theta) = k^{-1}\sum_l (2l+1)e^{i\delta_l}\sin\delta_l\, P_l(\cos\theta)$. In $\int(1-\cos\theta)|f|^2 \mathrm{d}\Omega$ the term with 1 gives $(4\pi/k^2)\sum_l(2l+1)\sin^2\delta_l$, and with $\cos\theta\, P_l = [(l+1)P_{l+1} + lP_{l-1}]/(2l+1)$ the term with $\cos\theta$ gives $(4\pi/k^2)\sum_l 2(l+1)\sin\delta_l\sin\delta_{l+1}\cos(\delta_l - \delta_{l+1})$. Their difference, with $\sin^2\delta_l + \sin^2\delta_{l+1} - 2\sin\delta_l\sin\delta_{l+1}\cos(\delta_l - \delta_{l+1}) = \sin^2(\delta_l - \delta_{l+1})$ and $(2l+1) = (l+1)+l$ regrouped, is Eq. (5.8).

B The integrals of the zero-range limit

(1) With $\hat{\jmath}_1(x) = \sin x/x - \cos x$ and the factor $e^{-\eta r}$,

$$\int_0^\infty \hat{\jmath}_1(k'r)\, r\cos(kr)\, e^{-\eta r}\mathrm{d}r = \frac{1}{k'}\int_0^\infty \sin(k'r)\cos(kr)e^{-\eta r}\mathrm{d}r - \int_0^\infty r\cos(k'r)\cos(kr)e^{-\eta r}\mathrm{d}r. \tag{B.1}$$

(2) The first integral is $\frac{1}{2k'}\int[\sin(k'+k)r + \sin(k'-k)r]e^{-\eta r}\mathrm{d}r \to \frac{1}{2k'}\left[\frac{1}{k'+k} + \frac{1}{k'-k}\right] = \frac{1}{k'^2-k^2}$.
(3) With $\int_0^\infty r\cos(qr)e^{-\eta r}\mathrm{d}r = (\eta^2-q^2)/(\eta^2+q^2)^2 \to -1/q^2$, the second integral tends to $-\frac{1}{2}\left[(k'-k)^{-2} + (k'+k)^{-2}\right] = -(k^2+k'^2)/(k'^2-k^2)^2$.
(4) The difference is $[(k'^2-k^2) + (k'^2+k^2)]/(k'^2-k^2)^2 = 2k'^2/(k'^2-k^2)^2$, which is Eq. (6.1). The integral with $\sin kr$ in place of $\cos kr$ tends to zero for $k \neq k'$, since $\int \sin ar \sin br\, e^{-\eta r}\mathrm{d}r$ and $\int r\sin(qr)e^{-\eta r}\mathrm{d}r = 2\eta q/(\eta^2+q^2)^2$ both vanish with η.

Table 8: Electron density in m^{-3} at which electron–ion and electron–atom absorption are equal at 1 THz, with the classical estimate of the thermal Gaunt factor and with the Born value [3]. Ions dominate above it.

fill	T (eV)	$\langle g\rangle_{\mathrm{cl}}$	$\langle g\rangle_{\mathrm{Born}}$	n_e, classical	n_e, Born
Ne	0.0259	1.00	1.58	7.0×10^{19}	4.5×10^{19}
Ne	0.1	1.00	2.25	9.9×10^{20}	4.4×10^{20}
Ne	0.3	1.28	2.83	9.1×10^{21}	4.1×10^{21}
Ne	0.63	1.89	3.23	3.3×10^{22}	2.0×10^{22}
Ne	1.08	2.34	3.52	9.3×10^{22}	6.2×10^{22}
Ar	0.0259	1.00	1.58	1.5×10^{20}	9.4×10^{19}
Ar	0.1	1.00	2.25	5.0×10^{20}	2.2×10^{20}
Ar	0.3	1.28	2.83	3.4×10^{21}	1.6×10^{21}
Ar	0.63	1.89	3.23	2.5×10^{22}	1.5×10^{22}
Ar	1.08	2.34	3.52	1.1×10^{23}	7.5×10^{22}

C The numerical checks

(1) A check script computes every boxed result of these notes and reproduces every number printed in them, from the tables it writes and from the prose. It needs numpy, scipy and the LXCat cross sections, runs about three minutes, and stops with an error if any check fails. The model potentials are fitted by a second script, and the check script verifies that the stored parameters reproduce the fit quality and the scattering lengths of Table 3.
(2) It uses four methods. Identities are verified against independent evaluations, the golden-rule cross section against the plane-wave Born formula and the Born mean-momentum form to 10^{-14}. Limits are verified by their convergence, the zero-range form by narrowing the well and the soft-photon and Fokker–Planck limits by the order of the remainder. The numerics are verified by halving the steps and doubling the ranges. The tables and the numbers in the text are printed from the same computation, and the script recomputes each number of the prose to the digits shown.

Table 9: The numerical checks of these notes, by the equation, table or section they verify. The second column counts all checks of the row, the third gives the largest deviation among its quantitative checks, relative except for the slopes of Eq. (5.9), which are absolute, the fourth counts the numbers printed in these notes that the script reproduces to their last digit, and the fifth the logical conditions.

Equation	Checks	Largest deviation	Printed	Logical
Eq. (3.11)	4	2.3×10^{-15}		
Eq. (3.6)	8	4.4×10^{-16}		
Eq. (6.1)	4	4.4×10^{-10}		
Eq. (6.2)	4	1.1×10^{-2}		1
Eq. (6.3)	3	4.7×10^{-16}		
Eq. (6.4)	3	1.2×10^{-16}		
Eq. (4.1)	8	8.5×10^{-14}		

Equation	Checks	Largest deviation	Printed	Logical
Eq. (4.4)	3	1.4×10^{-7}		
Eq. (4.9)	6	6.7×10^{-16}		
Eq. (4.7)	4	6.6×10^{-5}		1
Eq. (5.6)	4	9.2×10^{-5}		
Eq. (5.8)	6	5.1×10^{-8}		
Eq. (5.9)	12	5.9×10^{-3}		6
Eq. (5.12)	4	6.9×10^{-5}		
Eq. (6.7)	4	1.6×10^{-15}		
Eq. (6.8)	6			6
Sec. 7	9	2.0×10^{-16}	4	1
Table 2	1			1
Sec. 8	50	2.6×10^{-5}	6	14
Sec. 8.2	6	1.4×10^{-5}	3	1
Sec. 8.3	20	2.2×10^{-4}	8	6
Table 4	5			5
Table 5	1			1
Sec. 9	22		15	7
Sec. 10	14		10	4
Sec. 1	11		11	
Sec. 2	5		5	
Sec. 2.3	1			1
Sec. 3	1			1
Sec. 6.2	1			1
Sec. 5.1	2		1	1
Sec. 5	2		2	
Sec. 6.1	2		2	
Sec. 6.3	2		2	
Sec. 8.4	4		3	1
Total	242		72	59

References

[1] S. Atzeni and J. Meyer-ter-Vehn, *The Physics of Inertial Fusion* (Oxford University Press, Oxford, 2004), doi:10.1093/acprof:oso/9780198562641.001.0001.

[2] H. A. Kramers, Phil. Mag. **46**, 836 (1923), doi:10.1080/14786442308565244.

[3] W. J. Karzas and R. Latter, Astrophys. J. Suppl. **6**, 167 (1961), doi:10.1086/190063.

[4] F. V. Bunkin and M. V. Fedorov, *Bremsstrahlung in a strong radiation field*, Sov. Phys. JETP **22**, 844 (1966), record at http://www.jetp.ras.ru/cgi-bin/e/index/e/22/4/p844?a=list.

[5] H. Margenau, Phys. Rev. **69**, 508 (1946), doi:10.1103/physrev.69.508.

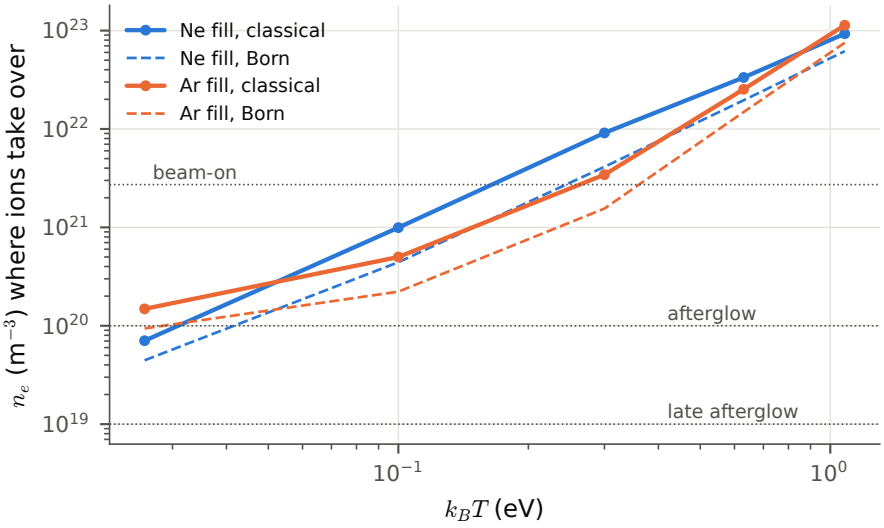


Figure 4: The electron density above which electron–ion absorption exceeds electron–atom absorption at 1 THz. Solid lines use the classical estimate of the Gaunt factor, dashed lines the Born value. The horizontal lines mark the electron density of the laser gas with the beam on, 2.7×10^{21} m^{-3}, and in the afterglow, 10^{20} and 10^{19} m^{-3}.

[6] V. Kas'yanov and A. Starostin, *On the theory of bremsstrahlung of slow electrons on atoms*, Sov. Phys. JETP **21**, 193 (1965), record at http://www.jetp.ras.ru/cgi-bin/e/index/e/21/1/p193?a=list.

[7] F. E. Low, Phys. Rev. **110**, 974 (1958), doi:10.1103/physrev.110.974.

[8] T. H. Burnett and N. M. Kroll, Phys. Rev. Lett. **20**, 86 (1968), doi:10.1103/physrevlett.20.86.

[9] N. M. Kroll and K. M. Watson, Phys. Rev. A **8**, 804 (1973), doi:10.1103/physreva.8.804.

[10] A. Dalgarno and N. F. Lane, ApJ **145**, 623 (1966), doi:10.1086/148801.

[11] T. Ohmura and H. Ohmura, ApJ **131**, 8 (1960), doi:10.1086/146799.

[12] S. Geltman, J. Quant. Spectrosc. Radiat. Transf. **13**, 601 (1973), doi:10.1016/0022-4073(73)90019-8.

[13] K. L. Bell and K. A. Berrington, J. Phys. B: At. Mol. Phys. **20**, 1329 (1987), doi:10.1088/0022-3700/20/6/017.

[14] R. L. Taylor and G. Caledonia, J. Quant. Spectrosc. Radiat. Transf. **9**, 657 (1969), doi:10.1016/0022-4073(69)90013-2.

[15] P. Ranson, O. Vallée and J. Chapelle, Rev. Phys. Appl. (Paris) **12**, 1829 (1977), doi:10.1051/rphysap:0197700120120182900.

[16] A. Buzulutskov *et al.*, Astropart. Phys. **103**, 29 (2018), doi:10.1016/j.astropartphys.2018.06.005.

[17] A. Weingartshofer *et al.*, Phys. Rev. Lett. **39**, 269 (1977), doi:10.1103/physrevlett.39.269.

[18] B. Wallbank and J. K. Holmes, Phys. Rev. A **48**, R2515 (1993), doi:10.1103/physreva.48.r2515.

[19] B. Wallbank and J. K. Holmes, J. Phys. B: At. Mol. Opt. Phys. **27**, 5405 (1994), doi:10.1088/0953-4075/27/21/027.

[20] S. Geltman, Phys. Rev. A **51**, R34 (1995), doi:10.1103/physreva.51.r34.

[21] T. F. O'Malley, L. Spruch and L. Rosenberg, J. Math. Phys. **2**, 491 (1961), doi:10.1063/1.1703735.

[22] P. Schwerdtfeger and J. K. Nagle, Mol. Phys. **117**, 1200 (2019), doi:10.1080/00268976.2018.1535143.

[23] M. Kitajima *et al.*, J. Phys.: Conf. Ser. **635**, 012030 (2015), doi:10.1088/1742-6596/635/1/012030.

[24] M. Kurokawa *et al.*, Phys. Rev. A **84**, 062717 (2011), doi:10.1103/physreva.84.062717.

[25] S. J. Buckman and J. Mitroy, J. Phys. B: At. Mol. Opt. Phys. **22**, 1365 (1989), doi:10.1088/0953-4075/22/9/010.

[26] T. F. O'Malley and R. W. Crompton, J. Phys. B: At. Mol. Phys. **13**, 3451 (1980), doi:10.1088/0022-3700/13/17/023.

[27] A. E. S. Green, D. L. Sellin and A. S. Zachor, Phys. Rev. **184**, 1 (1969), doi:10.1103/physrev.184.1.

[28] G. J. M. Hagelaar and L. C. Pitchford, Plasma Sources Sci. Technol. **14**, 722 (2005), doi:10.1088/0963-0252/14/4/011.

[29] P. A. G. Scheuer, Mon. Not. R. Astron. Soc. **120**, 231 (1960), doi:10.1093/mnras/120.3.231.

[30] L. C. Pitchford *et al.*, Plasma Process. Polym. **14**, 1600098 (2016), doi:10.1002/ppap.201600098.